

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1
PAPER 1 November 2019
2 hours 30 minutes

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. Evaluate $6 + 6 \times 4 - 2$. [3]
2. Evaluate $-11 - (-6)$. [3]
3. Work out $\frac{3}{5} + \frac{1}{4}$. Give your answer as a mixed number if possible. [3]
4. Find 25% of 250. [3]
5. Solve $5x + 7 = 42$. [3]
6. Share \$28 in the ratio 4 : 3. [4]
7. Find (a) the HCF and (b) the LCM of 24 and 20. [4]
8. Write 400000 in standard form. [3]
9. Simplify $5^4 \times 5^4$. Leave your answer as a power of 5. [3]
10. $n(A) = 15$, $n(B) = 12$ and $n(A \cap B) = 3$. Find $n(A \cup B)$. [3]
11. Expand and simplify $4(x + 4) - 6x$. [3]
12. Factorise completely $36x + 18$. [3]
13. Factorise $x^2 + 5x + 6$. [4]
14. Make h the subject of the formula $A = 2\pi r(r + h)$. Take π as a symbol. [4]
15. Solve $4x + 1 < 13$. [3]
16. Solve the simultaneous equations
$$\begin{aligned}x + y &= 7 \\x - y &= 5.\end{aligned}$$
[4]
17. The n th term of a sequence is $5 + 3(n - 1)$. Find the 8th term. [3]
18. In triangle ABC, angle A = 64° and angle B = 36° . Find angle C. [3]
19. A regular polygon has 6 sides. Find the size of each exterior angle. [3]
20. A right-angled triangle has shorter sides 18 cm and 24 cm. Calculate the hypotenuse. [4]
21. In right-angled triangle XYZ, angle X = 60° and hypotenuse XZ = 12 cm. Calculate XY, the side adjacent to 60° . [4]

- 22.** A triangle has base 12 cm and perpendicular height 7 cm. Find its area. **[3]**
- 23.** Calculate the area of a circle of radius 21 cm. Take $\pi = 22/7$. **[4]**
- 24.** A cuboid measures 7 cm by 3 cm by 7 cm. Find its volume. **[3]**
- 25.** Find the gradient of the line joining (4, 5) and (8, 13). **[3]**
- 26.** Find the midpoint of the line joining (4, 5) and (8, 13). **[3]**
- 27.** A bag contains 7 red, 3 blue and 3 green beads. A bead is drawn at random. Find the probability that it is red. **[3]**
- 28.** Find the mean of 8, 9, 5, 9, 4. **[3]**
- 29.** Work out $\begin{pmatrix} 1 & 0 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ where $\begin{pmatrix} & \\ & \end{pmatrix}$ denotes a 2×2 matrix times a column vector. **[4]**
- 30.** A trader in Mbare buys a crate of tomatoes for \$40 and sells it for \$45. Calculate the percentage profit. **[4]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).